

Bayesian nonparametric glottal inverse filtering

The problem

Speech is locally the convolution of the *glottal wave* and the *vocal tract filter*.

We can measure speech directly, but not its constituents.

Estimating both from speech alone is severely underdetermined: *glottal inverse filtering*, open for 50+ years.

It matters for early diagnosis of neurodegenerative disease and for biomarking of heart failure, which interfere with speech production early on.

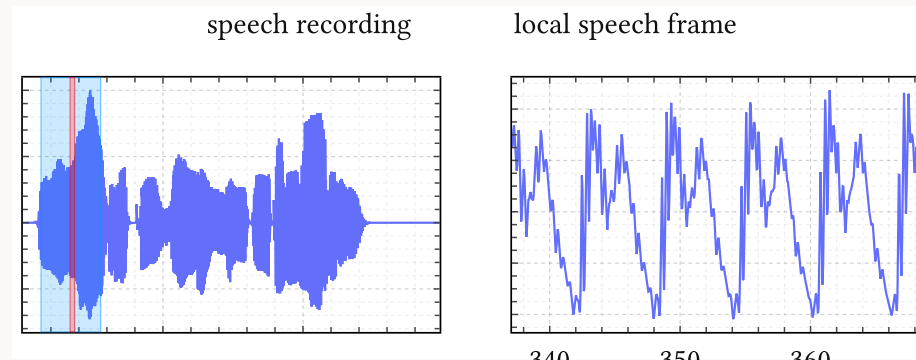
Our solution

Recast the problem as Bayesian inference: *Bayesian nonparametric glottal inverse filtering* (BNGIF).

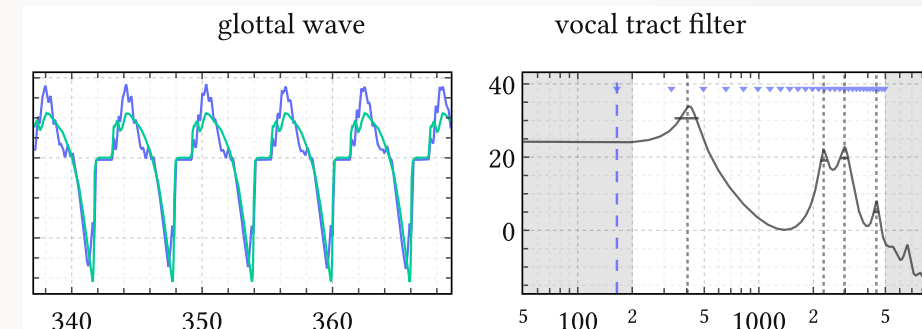
- A Gaussian process prior over glottal waves, learned from expensive physics-based simulations.
- The corpus, embedded and modelled as a Gaussian mixture: cheap surrogate models.
- The surrogates are the prior of a fast variational method. State of the art on the EGIFA/speech benchmark.

Marnix Van Soom · AI Lab, Vrije Universiteit Brussel
marnix@ai.vub.ac.be · mvsoom.github.io/bngif

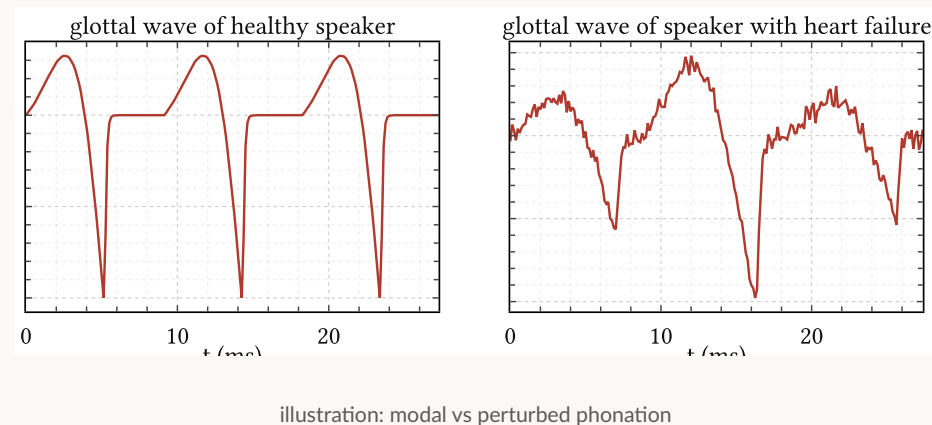
Speech signal



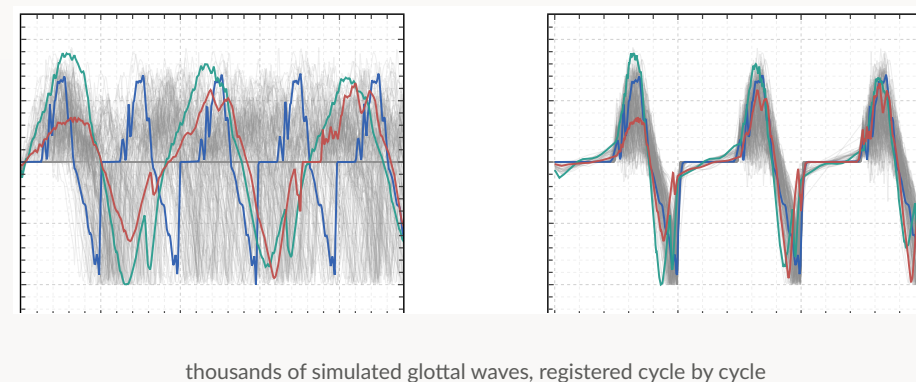
Glottal wave & vocal tract filter



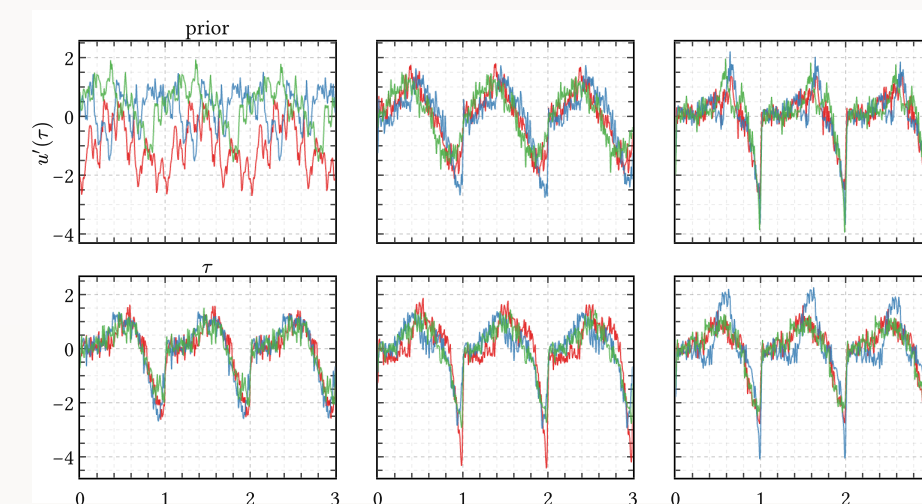
Speech diagnostics



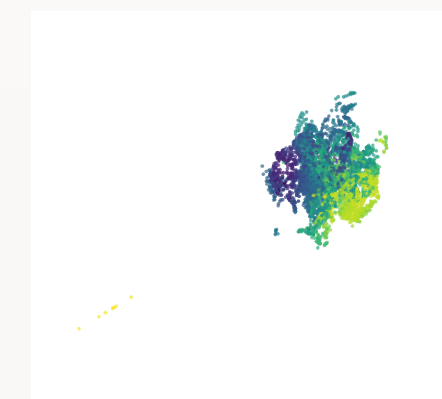
A corpus of glottal waves



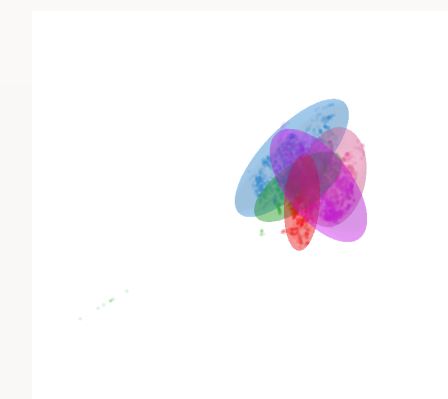
Corpus and mixture model



archetypes of the corpus, with samples



embedded corpus; each point is a glottal wave



mixture model of the embedded corpus